

MOOSONEE, ON, Canada

WMO: 713980

Lat: 51.291N

Lon: 80.617W

Elev: 9

StdP: 101.22

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -34.0	(c) -31.5	(d) -38.2	(e) 0.1	(f) -33.9	(g) -35.6	(h) 0.1	(i) -31.2	(j) 9.8	(k) -12.3	(l) 9.1	(m) -13.8	(n) 2.2	(o) 250	(p) 0.712

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 11.9	(c) 29.2	(d) 20.1	(e) 27.0	(f) 19.2	(g) 24.8	(h) 18.3	(i) 21.8	(j) 26.6	(k) 20.4	(l) 24.9	(m) 19.2	(n) 23.5	(o) 4.9	(p) 210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
20.1	14.8	24.2	18.8	13.6	23.0	17.5	12.5	21.8	63.9	26.9	59.0	24.8	55.0	23.6	26.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 9.4	(b) 8.3	(c) 7.3	(e) -37.6	(f) 33.0	(g) 2.8	(h) 1.6	(i) -39.6	(j) 34.1	(k) -41.3	(l) 35.1	(m) -42.9	(n) 36.0	(o) -44.9	(p) 37.1		
(a) 9.4	(b) 8.3	(c) 7.3	(e) -37.7	(f) 23.7	(g) 2.8	(h) 1.4	(i) -39.7	(j) 24.7	(k) -41.4	(l) 25.5	(m) -42.9	(n) 26.2	(o) -45.0	(p) 27.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	0.3	-18.5	-17.2	-10.6	-2.0	6.5	12.8	16.3	15.7	11.4	4.9	-3.3	-13.1
	DBStd	13.70	7.72	6.99	8.02	6.18	5.75	5.46	4.17	4.12	4.73	4.51	6.19	7.83
	HDD10.0	4150	883	763	640	362	144	31	1	3	37	172	399	715
	HDD18.3	6649	1142	996	898	611	369	183	92	104	214	418	649	973
	CDD10.0	620	0	0	0	1	36	114	197	178	80	13	1	0
	CDD18.3	77	0	0	0	0	3	16	29	21	7	1	0	0
	CDH23.3	783	0	0	0	1	54	209	278	188	52	2	0	0
	CDH26.7	213	0	0	0	0	14	69	78	43	8	0	0	0
Wind	WSAvg	3.5	3.4	3.5	3.6	3.7	3.7	3.5	3.3	3.3	3.5	3.6	3.7	3.5
Precipitation	PrecAvg	724	38	28	34	44	56	75	99	82	84	76	65	40
	PrecMax	1111	76	60	98	97	104	179	193	180	176	172	145	110
	PrecMin	540	9	4	7	11	12	15	19	30	37	28	24	10
	PrecStd	147	19	15	21	21	25	38	43	36	31	31	27	19
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	2.5	4.0	11.7	19.0	28.7	32.1	31.8	30.1	27.7	22.5	13.4	4.3
		MCWB	1.7	1.9	7.2	10.1	18.2	21.5	21.9	21.7	19.8	16.3	9.8	2.5
	2%	DB	0.2	0.5	7.1	13.3	24.0	28.7	28.8	27.8	24.7	17.9	8.8	2.2
		MCWB	-0.4	-0.5	3.6	7.2	15.2	18.8	20.3	20.3	18.6	13.8	6.1	1.3
	5%	DB	-2.2	-2.7	4.5	9.8	20.4	25.6	26.6	25.4	22.0	14.6	6.4	0.3
		MCWB	-2.8	-3.6	2.0	5.4	13.3	17.8	19.4	19.2	17.2	11.7	4.7	-0.3
	10%	DB	-6.0	-5.7	1.7	6.7	16.9	22.6	24.1	23.0	19.4	11.6	4.2	-1.6
		MCWB	-6.6	-6.4	-0.1	3.5	11.8	16.4	18.5	18.2	15.7	9.2	3.0	-2.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	1.9	2.3	8.3	10.8	19.4	22.9	23.3	23.0	21.5	17.0	10.3	3.3
		MCDB	2.4	3.5	10.1	16.4	25.9	29.1	29.4	27.5	26.2	20.3	12.5	4.0
	2%	WB	-0.2	-0.5	4.1	7.9	16.1	20.7	21.7	21.4	19.4	14.5	6.7	1.4
		MCDB	0.3	0.7	6.7	12.2	21.6	25.5	26.4	26.0	23.3	17.1	8.0	1.9
	5%	WB	-2.8	-3.6	2.1	5.6	14.1	18.9	20.3	20.0	17.8	11.9	4.7	-0.2
		MCDB	-2.3	-2.8	4.1	9.5	19.4	24.0	24.6	24.3	21.6	14.5	6.3	0.3
	10%	WB	-6.6	-6.4	0.0	3.6	12.0	17.1	19.1	18.6	16.0	9.4	3.1	-2.2
		MCDB	-6.1	-5.7	1.7	6.5	16.7	21.9	23.4	22.4	19.1	11.3	4.2	-1.6
Mean Daily Temperature Range	5% DB	MDBR	10.6	12.1	12.7	10.8	11.5	12.9	11.9	11.0	10.2	7.5	6.7	8.7
		MCDBR	10.8	13.4	13.4	15.3	18.3	16.6	14.7	14.0	13.8	12.0	8.7	8.1
		MCWBR	10.5	12.4	10.4	10.1	10.5	9.1	8.0	7.9	8.7	8.4	7.0	7.8
	5% WB	MCDBR	10.9	13.0	12.4	14.0	16.3	14.5	13.1	12.6	12.4	11.2	8.1	8.1
		MCWBR	10.6	12.2	10.0	9.8	10.1	9.3	8.1	7.9	8.6	8.6	6.8	7.9
Clear-Sky Solar Irradiance	taub	0.260	0.273	0.288	0.351	0.370	0.376	0.399	0.382	0.366	0.310	0.299	0.257	
	taud	2.227	2.229	2.254	2.208	2.295	2.373	2.357	2.406	2.430	2.560	2.415	2.276	
	Ebn at Noon	758	857	913	884	883	879	850	848	821	820	717	705	
	Edh at Noon	68	90	107	128	123	115	116	105	92	67	58	55	
All-Sky Solar Radiation	RadAvg	1.23	2.40	3.99	5.12	5.06	5.78	5.30	4.34	3.08	1.81	0.95	0.82	
	RadStd	0.14	0.16	0.29	0.50	0.43	0.45	0.37	0.32	0.23	0.20	0.10	0.12	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	+1.49	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (0 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page